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Community structure of metazoan parasites of the freshwater eel, *Macrognathus aculeatus* Bloch, 1786 from river Godavari, India

[1]ANU PRASANNA VANKARA AND [2]VIJAYALAKSHMI CHIKKAM

[1]Department of Animal Sciences, Yogi Vemana University, Kadapa, A.P-516 003
[2]Department of Zoology, Andhra University, Visakhapatnam-530 003

ABSTRACT - Five hundred and sixty one *Macrognathus aculeatus* collected from river Godavari during 2005-2007 were examined for the presence of metazoan parasites. 386 (68%) fishes were parasitized by at least one or more metazoan parasite species. Eight species of parasites were collected: 4 Digenea, 1 Monogenea, 1 Copepoda, 1 Nematoda and 1Isopoda. Endoparasites predominated the majority of the components of the infracommunities analyzed and represented 70.5% of the total parasites collected. The digenean, *Allocreadium aculeatum* was most prevalent in the parasite community of *M. aculeatus* but occupies the position of satellite species. *M. bam* occupies the position of secondary species. The metazoan parasite species, *Allocreadium aculeatum*, *Tetracotyle* larva and *M. bam* exhibited over-dispersed distributions within the fish population. Parasitization increased with increase in host size. Host sex had no influence on overall parasitization. However, individual parasites, *L. cyprinacea mastacembeli* and *C. mastacembeli* from *M. aculeatus* presented a significant impact in relation to sex as females showed relatively higher parasitization by these parasites.

KEYWORDS – Parasite ecology, Community structure, *Macrognathus aculeatus*, River Godavari.

Introduction

Macrognathus aculeatus Bloch, 1786 is the freshwater spiny eel of the genus *Mastacembelus* belonging to the family Mastacembelidae. It is a popular indigenous fish with a lot of economic and medicinal importance. This fish are attributed as highly proteinaceous food and is the delicious fish of southern India. It is commonly known as 'lesser spiny-eel or 'peacock-eel'. It spends most of its time at the bottom and conceals itself in the mud and hence, is also commonly known as "mud eel" and "mudi bommidai" locally in Andhra Pradesh and harbours a variety of metazoan parasitic fauna which includes monogeneans, digeneans, nematodes, copepods and isopods. Many research workers have dealt with taxonomic aspects of various groups of metazoan parasites of Mastacembelid fishes across the world (Harshey, 1937; Karve and Naik, 1951; Tripathi, 1959; Verma, 1973, Shinde and Jadhav, 1980; Agarwal and Kumar, 1981; Venkatanarsaiah, 1981; Agarwal and Agarwal, 1988; Agarwal and Sharma, 1989; Deshmukh and Shinde, 1989; Jadhav *et al.*, 1990; De and Dey, 1992; Dubey *et al.*, 1992; Noopur *et al.*, 1994; Wongsawad and Jadhav, 1998; Kritsky *et al.*, 2004; Jalali and Barzegar, 2006; Jalali *et al.*, 2008; Vankara and Chikkam, 2010).

Corresponding author: annuprasanna@gmail.com

The structure of metazoan parasite community of fish is now receiving major attention of ecologists. Parasitic communites is a complex of individuals belonging to different parasite species inhabiting a host (Cloutman, 1975). Study on seasonal dynamics of parasitism levels serves as a tool to understand the factors that affect the population biology of the host-parasite system (Chubb, 1982). Vast majority of work has been focused on the metazoan parasitic communities in various fishes including both ectoparasites and endoparasites (Thoney 1991, Aho and Bush 1993, Campos and Carbonell 1994, Molloy *et al*., 1995, Luque *et al*., 1996a, 1996b, Alves and Luque 2001a, 2001b, Aloo 2002, Alves *et al*., 2002, Guidelli *et al*., 2003, Aguilar-Aguilar and Salgado-Maldonado 2006 and Hernandez *et al*., 2007, Anu prasanna *et al*., 2010). However, the studies on the populational and ecological features of metazoan parasitic fauna of Mastacembelid fishes are very meager (Anu prasanna *et al*., 2010). Hence in this report, we analyzed the metazoan parasite community of Mastacembelid fish, *Macrognathus aculeatus* at component and infracommunity levels.

Material and methods

For the present study, 561 *Macrognathus aculeatus* were examined from August 2005 to September 2007. Fish were procured from the River Godavari, Rajahmundry and local fish markets in and around the river and were brought to the laboratory and examined for the presence of parasites. Fishes were identified according to Jayaram (1981), Munro (1982) and Day (1994). Length, weight and sex of each fish individually were noted carefully and all organs were examined separately for the collection of parasites. Parasites were collected, enumerated and permanent slides were prepared by fixing the parasites in A.F.A (Alcohol-85ml; Formaldehyde-10ml and Acetic acid-5ml) and stained with Alum carmine (Hiware *et al*., 2003 and Madhavi *et al*., 2007). Following Bush *et al*., (1997), prevalence, intensity and abundance were calculated for the parasites of the fish to know the core and satellite species among them. The quotient between variance and mean of parasite abundance is used to determine distribution patterns. The dominance frequencies of each parasite infracommunity were calculated by Berger-Parker index of dominance (Magurran 1998). Grabda-Kazubski *et al*. (1987) classified the nature of infection by parasite species into dominant (>70%), sub-dominant (50-70%), common (30-50%), frequent (10-30%), rare (4-10%) and sporadic (<4%) based on the percentage of prevalence. Pearson's correlation coefficient *r* was used as an indication of the relationship between the host's length and prevalence of parasites. The effect of host sex on abundance and prevalence of parasites was tested by applying chi-square test. Parasite species diversity was calculated using Shannon-Weiner index (H'). For each infracommunity, the evenness (Shannon-based evenness index, E) was calculated. Diversity H' and evenness (E) were calculated only for those fish infected with more than one parasitic species (diversity not defined for K=1). The pairwise associations were calculated for the three metacercarial stages, *Ascocotyle nana*, *Tetracotyle sp.* and *Clinostomum mastacembeli* in *M. aculeatus* by using Jaccard's coefficient (r_j) and modified Sørenson's coefficient (Poulin & Valtonen, 2001).

Results

Component community

Eight species of metazoan parasites (4 Digenea, 1 Monogenea, 1 Copepoda, 1 Nematoda and 1Isopoda) were recorded from the fish host, *M. aculeatus*. Table 1 shows the parasites found in the host species, their location and parasitological indices. Digeneans are the most abundant with four species accounting for 68% of total parasites collected. 246 hosts (43.85%) of *M. aculeatus* showed infection with any single parasite groups, 131 (23.35%) with any two parasitic groups and 9 (1.60%) with three parasitic groups (Table-2). No single fish was infected with all the parasitic groups simultaneously. Of the four digeneans, three are the larval stages of the avian trematodes, *Ascocotyle* (4%),

TABLE-1. Diversity parameters and distribution patterns of parasitic species infecting *Macrognathus aculeatus*

Name of the parasite	Infected fishes	No. of parasites	Prevalence	Mean intensity	Mean abundane	Index of infection	Range	D.I	Mean parasite richness	Location	Nature of infection *	Nature of species **
Allocreadium aculeatus	168	4836	28.3	13±9.5	8.2±9.3	5.4	1-154	0.473	8.62	Intestine	Frequent	Satellite
Tetracotyle metacercaria	72	1105	13.2	5.25±2.27	2.0±2.97	0.96	2-62	0.11	1.97	Body cavity	Frequent	Satellite
Ascocotyle metacercaria	119	452	20.2	2.6±1.2	0.81±1.3	0.3	1-10	0.044	0.81	Gills	Frequent	Satellite
Clinostomum metacercaria	130	816	23.7	3.69±2.03	1.45±2.14	0.73	1-24	0.079	1.45	Body cavity	Frequent	Satellite
Mastacembelocleidus bam	243	2251	42.3	7.7±3.71	3.9±5.09	2.2	1-68	0.220	4.01	Gills	Common	**Secondary**
Lerneae cyprinacea mastacembeli	129	722	21.7	2.99±1.81	1.27±1.75	0.57	1-26	0.068	1.28	Gills	Frequent	Satellite
Alitropus typus	29	43	5.25	0.65±0.25	0.077±0.2	0.01	1-3	0.004	0.077	Skin	Rare	Satellite
Camallanus unispiculus	2	2	0.35	1±0	0.0035±0	0.00001	0-1	0.0002	0.0035	Intestine	Sporadic	Satellite

Clinostomum (7%) and *Tetracotyle sp.* (10%) and one adult digenean, *Allocreadium aculeatum* which was the dominant species, with 4836 parasites (Range=1-154) collected (47.3% of total parasites) with a mean of 9 parasites/fish and shows the highest values of mean relative dominance of 0.473 ± 0.0247 (Table-1). Only a single species of monogenean, *Mastacembelocleidus bam* (22% of all parasites) collected, occupies the next dominant position in the parasite assemblages. *Lernaea cyprinacea mastacembeli* collected from the gills of the host was of very sporadic occurrence; however, its copepodid stages were recurrently collected from the host during the study period and this copepod together with its copepodid stages comprised 7% of the total parasites. Only 43 isopod parasites (less than 1% of total parasites collected) were obtained from the skin of fish. Although only two nematodes were collected during the study period, they were considered in the community studies. The ratio of variance to mean values gives the index of dispersion. *Allocreadium aculeatum*, *M. bam* and *Tetracotyle* larva exhibited over-dispersed distributions within the fish population (Table-5).

Infracommunities

Three hundred eighty-six (68%) fishes were parasitized by at least one or more parasite species. A total of 10227 individual parasites were collected with a mean of 18 parasites/fish. One hundred fourteen hosts (20%) showed infection with one parasite species and 109 (19%), 101 (18%), 50 (9%) and 12 (2%) had multiple infections with 2, 3 and 4 parasite species respectively (Table-4). Shannon's H' index and Shannon-based evenness index (E) values are given in Table-6. In the present study, most of the parasitic species viz. *Allocreadium aculeatum*, *L. cyprinacea*, metacercaria *Ascocotyle nana*, metacercaria *Tetracotyle sp.* are common with a prevalence ranging between 10-30% in the hosts. The monogenean, *Mastacembelocleidus bam* was found commonly

TABLE-2. Frequency distribution of number of parasitic groups per individual in *Macrognathus aculeatus*

Sl. No.	No. of parasitic groups	No. of infected fishes	% of frequency
1	1	246	43.8
2	2	131	23.35
3	3	9	1.60
4	0	0	0

n = 386, Óx = 3, X = 3/386 = 0.007, Range = 1-3

TABLE-3. Number of parasites obtained, Dominance index and mean total parasites of different parasitic groups in *M. aculeatus*

Sl. No.	Parasite group	No. of parasites	Dominance index	Mean total parasites
1	Monogeneans	2251	0.220	4.01
2	Digeneans	7209	0.704	12.8
3	Cestodes	0	0	0
4	Nematodes	2	0.0002	0.0035
5	Copepods	722	0.071	1.29
6	Isopods	43	0.0042	0.077

TABLE-4. Frequency distribution of number of parasitic species per individual in *M. aculeatus*

Sl. No.	No. of parasitic species	No. of infected fishes	% of frequency
1	1	114	20.32
2	2	109	19.4
3	3	101	18
4	4	50	8.9
5	5	12	2.13
6	6	0	0
7	7	-	-

infecting 30-50% of *M. aculeatus*. Parasitisation by the Isopod, *Alitropus typus* was rare (4-10%), whereas *C. unispiculus* show sporadic infection (<4%). There is no core species in the host. Only secondary and satellite species were encountered. Although the density of *Mastacembelocleidus bam* was less, its infection rate was high (i.e, number of infected hosts are more) and hence they occupy the position of secondary species, whereas *A. aculeatum* occurs in large numbers in the host but still occupies the position of satellite species due to their lower infection rate. However, mean intensity and mean abundance was higher for *A. aculeatum* than *M. bam.* The digenean, *Clinostomum mastacembeli* occupies third position followed by *L. cyprinacea mastacembeli* and metacercaria *Ascocotyle nana* respectively. Remaining parasitic species, *Tetracotyle sp.*, *Camallanus unispiculus* and Isopod, *Alitropus typus* occupy the position of satellite species (Table-1). Berger-Parker's dominance index and mean of total parasite species was calculated for each species of

TABLE-5. Comparison of the mean (X), Variance (s^2) and Dispersion index (s^2/X) of parasite species in *M. aculeatus*

Name of the parasite	2005-2006 (n =284)				2006-2007 (n = 277)			
	No. of parasites collected	Mean (X)	Variance (s^2)	Dispersion index (s^2/X)	No. of parasites collected	Mean (X)	Variance (s^2)	Dispersion index (s^2/X)
Mastacembelocleidus bam	1246	4.39	81.39	18.53	1005	3.628	34.87	9.6
Allocreadium aculeatum	2544	8.95	377.61	42.19	2292	8.27	457.22	55.28
Tetarcotyle metacercaria	458	1.61	37.77	23.45	647	2.33	54.16	23.24
Metacercaria of Ascocotyle nana	266	0.93	4.61	4.95	186	0.67	2.20	3.28
Clinostomum mastacembeli	450	1.58	13.63	8.62	366	1.32	10.38	7.86
Lernaea cyprinaceae	360	1.26	9.50	7.53	362	1.34	9.63	7.18

TABLE-6. Diversity parameters of metazoan parasite communities of *M. aculeatus*

Host	Sample size	Mean no. of parasite species	Mean no. of parasite individuals	Shannon's diversity index (H')	Shannon-based evenness (E)	No. of core species
Macrognathus aculeatus	561 (272)	3.08 ± 1.07 Range (2-6)	33.7 ± 26.60 Range (3-156)	0.78 ± 0.40 Range (0.02-3.16)	0.74 ± 0.35 Range (0.02-4.36)	-

TABLE-7. Matrix of pairwise associations between larval helminths using Jaccard's coefficient, r_j; each fish sample treated separately

Host species	*Tetracotyle sp.*	*Ascocotyle sp.*	*C. mastacembeli*
M. aculeatus			
Tetracotyle sp.	-	0.143	0.015
Ascocotyle sp.	-	-	0.210
C. mastacembeli	-	-	-

TABLE-8. Matrix of pairwise associations between larval helminths using modified Sorensen's coefficient, r_s; each fish sample treated separately

Host species	*Tetracotyle sp.*	*Ascocotyle sp.*	*C. mastacembeli*
M. aculeatus			
Tetracotyle sp.	-	-0.515	-0.95
Ascocotyle sp.	-	-	0.210
C. mastacembeli	-0.52	-	-

TABLE-9. Correlation of parasite richness, parasite diversity and evenness in the different size classes of *M. aculeatus*

Size class	Parasite richness	Parasite diversity (H')	Evenness (E)	Correlation coefficient (r)
9-12	5±0	0.92±0	0.575±0	0.088
13-16	2.5±0.77	0.72±0.28	0.81±0.24	
17-20	2.9±0.96	0.74±0.36	0.74±0.28	
21-24	3.34±1.12	0.77±0.35	0.68±0.25	
25-28	3.18±1.10	0.91±0.65	0.84±0.68	
29-32	4.4±1.14	1.072±0.28	0.74±0.17	

parasite in the host. *A. aculeatum* presents the highest dominance value of 0.473 followed by *Mastacembelocleidus bam* being 0.220 in *M. aculeatus*. Mean of total parasitic species of *A. aculeatum* and *M. bam* was 8.62 and 4.01 respectively (Table-1). Jaccard's coefficient of association between larval helminthes was low for 2 out of 3 pairwise combinations in *M. aculeatus* which indicates that the two species in a pair do not co-occur very often in the same fish (Table–7). The *Clinostomum–Tetracotyle* pair in *M. aculeatus* shows very negligible or almost no association ($r_{j\,=}$ 0.015; r_s = - 0.95). However, Sørensen's coefficient for all the pairwise combinations showed negative coefficient values suggesting that the two species in a pair do not co-occur recurrently in the same fish (Table-8). Host size serves as an important factor in determining the burden of parasitic communities in a host. *Macrognathus aculeatus* measured 9-31cm (mean = 19.71±3.72 cm) in total length and were categorized into 6 classes. Pearson's coefficient was 0.088 suggesting that *M. aculeatus* shows very less correlation between host size and parasite burden. Parasite species richness was correlated with total body length of fish (r = 0.480) and parasite diversity was also correlated with the host total body length (r = 0.141). The six most prevalent individual species (>10%) of parasites in *M. aculeatus* are, the monogenean, *M. bam* (r = 0.03), the digeneans, *A. aculeatum* (r = 0.11) *Ascocotyle nana* (r = 0.02), *Tetracotyle sp.* (r = 0.114), *Clinostomum mastacembeli* (r = - 0.015) and the copepod *Lernaea cyprinacea mastacembeli* (r = 0.202). Overall parasitization is low in small sized fish under 12cm, and increases from medium sized fish to larger fishes (Table-9). Similar pattern was observed for both males and females. Host sex was one of the biotic factors which play an important role in determining the parasitization in a host.

TABLE-10. Diversity parameters of parasitic species in males and females of *M. aculeatus*

Host name	***Macrognathus aculeatus*(N_m = 236 ; N_f = 281)**							
Parasite	**N_{mi}**	**N_{fi}**	**P_m**	**P_f**	**MI_m**	**MI_f**	**MA_m**	**MA_f**
Monogenea 1. *Mastacembelocleidus bam*	108	128	45.76	45.6	9.7	9.1	4.47	4.15
Digenea2. *Allocreadium aculeatum*	70	92	29.6	32.7	26.6	31.46	7.9	10.3
3. *Clinostomum* sp. (M)	51	76	21.6	27	5.5	6.8	1.18	1.85
4. *Ascocotyle sp.* (M)	52	63	22	22.4	3.81	3.76	0.84	0.84
5. *Tetracotyle sp.* (M)	29	41	12.3	14.5	14.7	16.2	1.8	2.3
Nematoda 6. *Camallanus unispivulus*	0	2	0	0.71	0	1	0	0.007
Copepoda 7. *L.cyprinancea mastacembeli*	47	77	19.9	27.4	6.06	5.53	1.2	1.51
Isopoda 8. *Alitropus typus*	11	17	4.66	6.04	1.45	1.47	0.067	0.088

*$N_{m=}$ Number of males examined; $N_{f=}$ Number of females examined; $N_{mi=}$ Number of males infected;
$N_{fi=}$ Number of females infected; P_m & P_f = Prevalence of males and females respectively;
MI_m & MI_f = Mean intensity of males and females; MA_m & MA_f = mean abundance of males and females respectively.

TABLE-11. Influence of seasons on parasitization of *Macrognathus aculeatus*

***Macrognathus aculeatus* (n=561)**

Seasons	**fishes examined (a)**	**No. of infected fishes (b)**	**Total parasites (c)**	**Prevalence (%)**	**Mean intensity**	**Mean abundance**	**Index of infection**
			2005-2006				
Winter (14°C)	103	88	3288	85.4	37.4	31.9	27.3
Summer (38°C)	90	54	419	60	7.8	4.7	2.8
Rainy (22°C)	95	66	1698	69.5	25.7	17.9	12.4
			2006-2007				
Winter (16°C)	99	76	2514	76.8	33.1	25.4	19.5
Summer (39°C)	86	41	332	47.7	8.1	3.9	1.8
Rainy (21°C)	88	61	1976	69.3	32.4	22.5	15.6

Of the total sample of 561 *M. aculeatus*, 281 are females (50%), 236 are males (42%) and 44 fish are juveniles (8%). Among these fish, 214 females (38%), 156 males (28%) and 15 with undetermined sex (3%) were parasitized by one or more parasite species. Impact of host sex on the overall prevalence of infection, estimated by chi square test was applied for each host species sex-wise individually. No statistically significant association (p=0) occurs between the parasite abundance in male and female fishes. *L. cyprinacea mastacembeli* and the *Clinostomum mastacembeli* (Metacercaria) appear to be slightly influenced by the sex of the host as these indices are higher in females than males (Table 10).

Discussion

Overall parasitization and community diversity of the host shows good consistency for consecutive years. *M. aculeatus* can be considered to be good intermediate or paratenic hosts since the parasitic fauna of these fishes are predominantly larval helminths. Larval helminths use fish as intermediate hosts for the completion of their life-cycles in the definitive vertebrate hosts (Luque and Poulin, 2004). There are studies on the adult helminths in the gastrointestinal tract of fish, few on ectoparasites since these endohelminths and ectoparasites form two distinct guilds of parasites inhabiting the same host and compete for the same resources (Agarwal and Kumar, 1981; Agarwal and Singh, 1982; Amato & Amato, 1993; Akenova, 2007). But larval helminths do not form an interactive guild as they appear less likely to compete for resources since they seek shelter in various organs and because many of them are in the form of cysts and do not actively feed on host nutrients or tissues. Parasitic infection shows a seasonal periodicity and cyclic variability for *M. aculeatus*. Parasitization was high during winter as compared to other two seasons for both the fishes (Table-11). However, the peak periods vary from fish to fish or group to group. High population density of adult and larval parasites during winter may be due to the large population of the intermediate host (*M. aculeatus*). The high population density of *A. aculeatum* in winter can be positively correlated with the high population density of the host, *M. aculeatus* in the winter months. Recruitment of the parasites may take place after summer and reach their peak periods in the winter months. Banu *et al*. (1993), Hossain *et al*. (1994), Akhter *et al*. (1997) and Chandra *et al*. (1997) also reported that incidence of diseases in fish is high during winter. The results of the present study are in good concurrence with these reports. Temperature is one of the crucial factors controlling parasitic infection (Hopkins 1959, Chubb 1963, Awachie 1966, Kennedy 1971, 1975, 1977a & 1977b, Muralidhar 1989, Rohde 1993, Rodrigues and Saraiva 1996, Chapman *et al*., 2000, Turner 2000, Mouristen and Poulin 2002). During winter, the fall in water temperature reduces the immune response in fish which make them more vulnerable to infection (Banu *et al*., 1993). Infection was almost negligible during summer in *M. aculeatus* during the present study which can be attributed to their process of undergoing aestivation during unfavorable conditions. Diet, feeding habits, vagility of host species are the main factors affecting the parasitic community structure especially for the parasites transmitted to their final host (Sasal *et al.*, 1997). In the present study food and feeding habits of the host may be the important factors for the larval parasitic abundance as *M. aculeatus* is carnivorous and its diet constitutes insects, crustaceans, molluscs, snails and worms which serve as primary intermediate hosts for most of the digeneans and cestodes. Many parasitic communities are known to occupy precise, predictable and limited locations (sites) within their hosts (Crompton, 1973; Holmes, 1973; Crompton and Nesheim, 1976). Holmes and Price (1986) suggested that the number of species of parasites that a host species supports widely varies from host to host which might be due to the biotic and abiotic factors. Various factors responsible for the parasite composition within a host is host size, phylogeny, composition of diet, complexity of gut and vagility. This is true in case of

M. aculeatus as these fishes occur in shallow waters where the movement is limited. Moreover, *M. aculeatus* shows low parasitic burden and poor diversity of parasites which might be due to the habit of undergoing aestivation during summer (dry and unfavourable conditions) thereby having reduced capacity of feeding. Kennedy and Bush (1994) are of the opinion that the examination of helminth community structure at the level of host's family and genus over a particular geographical area will improve our understanding of unpredictable helminth communities and their determinant factors. This statement is in total agreement with the present study. Values of J_r and J_s coefficients of pairwise interspecific association for *Tetracotyle*- *Ascocotyle*, *Ascocotyle*- *Clinostomum* and *Tetracotyle*- *Clinostomum* in *M. aculeatus* suggested that two species in a pair do not co-occur very often in the same fish individuals. The present study was also in accordance with the views of Poulin and Valtonen (2001) which suggests that assemblages of larval helminth parasites in fish are not random selections of locally available species, but rather non-random packets of larval parasites that travel together along common transmission routes. Few parasitic species are narrowly specific to one species of hosts which are called as 'monoxenic' or 'stenoadaptive' parasites whereas the parasites with wide specificity constitute 'polyxenic' or 'euryadaptive' species (Margolis *et al*., 1982). In the present study, *Allocreadium aculeatum* is a monoxenic species with narrow specificity for *M. aculeatus* whereas other parasitic species *M. bam*, *Ascocotyle nana, Clinostomum sp. Tetracotyle sp., Alitropus typus* and *Lernaea cyprinacea mastacembeli* are polyxenic species. At the level of parasite component community, the concept of core and satellite species concept plays a crucial role in explaining the species interactions (Holmes and Price, 1986; Esch *et al*., 1990; Sousa, 1994). In the present study, *A. aculeatum* population was highest during winter but occupies the position of secondary species in *M. aculeatus*. Helminth communites may be characterized as being 'isolationists' or 'interactive' and they are recognized only when the species regularly co-occur (Holmes and Price, 1986). The presence of *A. aculeatum* and *Mastacembelocleidus bam* restricts the presence of larval parasites during winter in *M. aculeatus* which suggests the process of antagonism being operated within the host system. Shotter (1973), Bell and Burt (1991), Machado *et al*. (1994), Poulin (1995), Fiorillo and Font (1996) and Luque *et al*. (1996a & 1996b) stated that helminth diversity and parasite community richness are more consistently correlated with host size. Takemoto and Pavanelli (1994) and Machado *et al*. (1994) found that increase in size and age of fish cause a significant increase in the levels of parasitism. Larger hosts may consume greater quantities of food and are exposed to a wider range of parasitic stages. Larger hosts offer more space to parasites and can also support greater number of species and the parasitization in *M. aculeatus* support these views. Thomas (1964) suggested that effects of host sex on levels of parasitism might be due to the physiological, biological and behavioral difference between males and females which produce a small albeit consistent sexual trend in infection levels. Females are more likely to be heavily infected when compared to males which might be due to the stress caused during reproductive periods and lead into behavioral changes, thus making them more vulnerable to heavy parasitization. In contrast, Folstad and Karter (1992) and Poulin (1996) found higher parasitization in males which might be due to high levels of testosterone which causes immunosuppression making them more susceptible to parasites than females. There are also studies by Lawrence (1970), Pennyquick, (1971a, b), Kennedy (1975), Muzzall (1980), Belghyti *et al*. (1994); Takemoto and Pavanelli (1994) and Machado *et al.* (1994), Luque *et al.* (1996a), Takemoto and Pavanelli (2000) and Lizama *et al*. (2005) who found host sex not to be a significant factor in determining the infection rate of helminth parasites in host fishes. The present study also supports the above studies since host sex did not influence overall parasitization. However, parasitization by *L.*

cyprinacea mastacembeli and *C. mastacembeli* was higher in female *M. aculeatus*. The present ecological studies suggest that the parasite communities of *M. aculeatus* are predictable, depauperate and non-interactive. Parasite diversity and richness can be attributed to its phylogeny and presence of intermediate hosts in the area. Abundance of parasite population is attributed to feeding capacity of the host which is governed by biotic and abiotic factors like temperature, water currents etc. The present study also holds well with the views of Holmes (1990) which suggest that fish as intermediate hosts have rich parasite fauna since they harbour both adults and larval helminths and that freshwater fish parasitic communities are less diverse than their marine counterparts.

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